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How open is innovation?

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ABSTRACT

This paper is motivated by a desire to clarify the definition of 'openness' as currently used in the literature on open innovation, and to re-conceptualize the idea for future research on the topic. We combine bibliographic analysis of all papers on the topic published in Thomson's ISI Web of Knowledge (ISI) with a systematic content analysis of the field to develop a deeper understanding of earlier work. Our review indicates two inbound processes: sourcing and acquiring, and two outbound processes, revealing and selling. We analyze the advantages and disadvantages of these different forms of openness. The paper concludes with implications for theory and practice, charting several promising areas for future research.

1. Introduction

How does openness influence firms' ability to innovate and appropriate benefits of innovation? These questions lie at the heart of recent research on innovation (e.g. Chesbrough, 2003a; Helfat, 2006; Laursen and Salter, 2006a). Their answers require a conceptual frame that defines and classifies different dimensions of openness. There has been a range of important papers published on the topic and it is timely to take stock on where the research stands to advance it further. Our review shows that a variety of definitions and focal points are used, but that these do not yet cohere into a useable analytical frame. The absence of such a framing device makes it difficult to compare and evaluate the advantages and disadvantages to openness at the level of the firm.[1]

A starting point for the idea of openness is that a single organization cannot innovate in isolation. It has to engage with different types of partners to acquire ideas and resources from the external environment to stay abreast of competition (Chesbrough, 2003a; Laursen and Salter, 2006a). This has stimulated questions about the role of openness in innovation that emphasizes the permeability of firms' boundaries where ideas, resources and individuals flow in and out of organizations. In this view, external actors can leverage a firm's investment in internal R&D through expanding opportunities of combinations of previously disconnected silos of knowledge and capabilities (Fleming, 2001; Hargadon and Sutton, 1997; Schumpeter, 1942). The downsides of openness can also be considerable, although there is less focus on this in the literature. Openness can result in resources being made available for others to exploit, with intellectual property being difficult to protect and benefits from innovation difficult to appropriate.

In defining openness, Chesbrough (2003a, p. XXIV) argues that "open innovation is a paradigm that assumes that firms can and should use external ideas as well as internal ideas, and internal and external paths to market, as firms look to advance their technology". Chesbrough's definition of openness, the most commonly used in the literature, is broad and underscores that valuable ideas emerge and can be commercialized from inside or outside the firm. The concept has common currency for at least four reasons. First, it reflects social and economic changes in working patterns, where professionals seek portfolio careers rather than a job-for-life with a single employer. Firms therefore need to find new ways of accessing talent that might not wish to be employed exclusively and directly. Second, globalization has expanded the extent of the market that allows for an increased division of labour. Third, improved market institutions such as intellectual property rights (IPR), venture capital (VC), and technology standards allow for organization to trade ideas. Fourth, new technologies allow for new ways to collaborate and coordinate across geographical distances.

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[1] We are aware that scholars have argued that open innovation can be explored at different levels of analysis including individuals, firms, sectors or even national innovation systems (West et al., 2006). For scholars interested in other units of analysis, we encourage reading West et al. (2006) and Christensen et al. (2005).

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In spite of rising interest in using the openness construct, systematic studies of openness remain cumbersome because of conceptual ambiguity. The extant literature presents the concept of openness in quite different ways; Laursen and Salter (2006a) equate openness with the number of external sources of innovation, whereas Henkel (2006) focuses on openness as revealing ideas previously hidden inside organizations. Our approach is to provide an analytical frame of different forms of openness and the associated advantages and disadvantages for each type.

To do so, we review the literature by an analysis of all papers published on open innovation in Thomson's ISI Web of Knowledge (ISI) to August 2009. Our starting point is a focal firm and the different forms of openness available to this organization. We systematically analyze the literature and its intellectual pillars by investigating the work that scholars cite. To get a sense of the community that has formed around this concept, we provide an overview of who has been working with whom in advancing the concept. After establishing these broad trends, we read all papers and categorized them in a systematic fashion. We develop an analytical frame by structuring the analysis in two dimensions: (1) inbound and (2) outbound (Gassmann and Enkel, 2006) versus (3) pecuniary and (4) non-pecuniary. This enables us to discuss two forms of inbound innovation—Acquiring and Sourcing; and two outbound—Selling and Revealing.

To date, the literature has been imbalanced in its strong focus on benefits of openness. Thus, we also pay close attention to disadvantages. We suggest that these factors might affect reasons why some firms gain and others lose from openness.

The paper is organized in six sections that combine to develop a framework based on prior conceptual and empirical work. The next section presents our method, followed by a review of the antecedents of openness found in literature on theories of the firm. Section four presents the different types of openness which emerged from our literature review. Our review enabled us to classify articles according to different focal points and these clustered according to the types that form the basis of our review. In section five we discuss implications for theory and practice, articulating promising areas for future research. The paper ends with concluding remarks about the study of open innovation.

2. Review method

We adopted an approach similar to the systematic reviews used in medicine in which systematic searches and formal summaries of the literature are used to identify and classify results of all major studies on a particular topic (Higgins and Green, 2006). We searched the ISI database for articles that had 'open innovation' in the topic field. The topic field includes the title, key words and abstract in the database. ISI is generally considered the most comprehensive database for scholarly work and includes thousands of journals. Although not all journals are included, ISI typically includes the most prominent journals in a field. This search yielded 701 papers that we downloaded to a local database in August 2009. We intentionally used a broad definition as the concept is used and published in a broad range of journals. We captured papers about how firms 'open up their innovation processes' and not only the specific term 'open innovation' or 'openness'. This broad search term, however, introduced papers that had little to do with the open innovation literature. We therefore read through all 701 abstracts to assess whether they dealt with open innovation. When we were unsure, we downloaded and read the full publication. This screening resulted in a short list of 150 papers where we read the full paper. While not a complete list of papers, the steps taken in data collection result in no bias towards any particular set of authors. Because ISI does not include books, we lack some important contributions to the field (such as Chesbrough's original book from 2003).[2] We opted for this approach to make it as transparent as possible. We make our list of papers available for other researchers to build upon (Email the corresponding author for a copy of the database).

ISI do not have a systematic approach to giving authors a unique identifier. There can be misspellings and middle initials that are not consistent between publications. One example includes Chesbrough who is listed with both Chesbrough, H. and Chesbrough, H.W. From this list of authors, we therefore disambiguated author names manually by looking at different variations of spelling, checking these against CVs. Having provided each author with a unique identifier, we produced tables and graphs of the most prolific authors, identifying patterns of co-authorship and the types of literature that the community draw upon. To illustrate how the community of scholars involved in open innovation evolves, we graphed the network using Pajek.

We read the papers to develop broad categories of how they treat the idea of openness, conceptually and empirically, to provide qualitative interpretation. We investigated how earlier literature used openness and the different forms of inbound and outbound innovation (Gassmann and Enkel, 2006; Van de Vrande et al., 2009). In this framework, inbound innovation refers to how firms source and acquire expertise and outbound to how firms attempt to sell ideas and resources in the marketplace. To further extend the literature, we discuss the associated advantages *and* disadvantages of each form of openness.

3. The boundaries of the firm and openness

Openness is in part defined by various forms of relationship with external actors and is thus closely coupled to a broader debate about the boundaries of the firm. In transaction cost economics the boundaries of the firm are given when it is difficult to anticipate all possible contingencies—and, by extension, to set prices. In these cases, interactions are assumed to be organized in firms rather than in the marketplace (Coase, 1937; Williamson, 1975). Transactions are distant interactions categorized by low levels of trust where conflicts are resolved by legal mechanisms. Markets offer flexibility in the choice of counterpart and tend to be swift and simple. Hierarchies within organizations are distinguished by employment relations, organizational structures and established norms.

Early work by Williamson and others has been followed by greater consideration of different hybrids between markets and hierarchies. For instance, Powell stresses how networks play a central role in knowledge creation in many fields (Powell, 1990). Williamson's more recent work also considers hybrid arrangements. Langlois (1992) develops a notion of 'dynamic transaction costs', referring to costs related to negotiating persuading and teaching potential partners with valuable resources as well as costs to those lacking resources when in need.[3] Langlois (2003) argues that managers must find new ways to conceptualize the 'post-Chandlerian firm' where innovation proceeds along less hierarchical lines because "large vertical integrated organizations are becoming less significant and are joining a richer mix of organizational forms" (Langlois, 2003, p. 353). From these viewpoints, the open innovation literature can be viewed as an instance of how firms make decisions whether to develop innovations internally or partner with external actors. External actors in this broad

[2] Books enter our analysis when we investigate the theoretical pillars of the field because the ISI database includes all references cited by the papers. We use this information to show the most cited papers in this stream of literature.

[3] We are grateful to an anonymous reviewer for encouraging us to advance the consideration of the boundaries of the firm.

sense could be either other organizations or individuals that are not employed by the focal firm.[4]

This calls for an appreciation of how internal R&D and openness are connected. There are several explanations as to why R&D can have an internal focus. Mowery (1983) explains how internal R&D emerged as a response to lower costs of organizing inside the firm compared to acquiring ideas and resources from the marketplace. Companies with significant investment in R&D can develop different organizational structures to streamline the innovative process. Firms can thus gain economies of scale and scope for their R&D (Henderson and Cockburn, 1996).

In spite of significant investments in R&D and strong internal resource endowments, it is sometimes important for firms to seek new resources outside their boundaries. Detailed empirical studies of innovation processes note that firms have always sourced from outside. In the late 19th century, Edison's laboratory—the Invention Factory at Menlo Park, displayed characteristics that in many regards had an open approach to innovation. The commercial development of electric lighting, for instance, was the product of a team of engineers that recombined ideas from previous inventions, collaborating with scientists, engineers, financiers and people in marketing outside the laboratory (Hargadon, 2003).

Firms remain investing in R&D despite their reliance on external partners. One explanation is Cohen and Levinthal's (1989) suggestion of a dual role of R&D: to develop new internally and to create the absorptive capacity to track and evaluate developments outside firm boundaries. They observed that firms with high investments in R&D appear to be more able to benefit from 'spill-overs' (Cohen and Levinthal, 1990).

Another aspect of maintaining internal expertise is Rosenberg's (1990) argument about conducting R&D as a 'ticket of admission' to potential partners. Firms with plentiful resources and expertise are more attractive partners. In the alliance literature, for instance, there are many detailed examples of how firms gain expertise through creating relationships with reputable partners (Powell et al., 2005).

To summarize, much of the literature views R&D as a necessary *complement* to openness for ideas and resources from external actors. It is less clear whether there could be a *substitution* effect, with openness replacing internal R&D. A potential substitution effect emerges where firms can streamline R&D and engage in external interactions to compensate for a more limited internal R&D focus (Chesbrough, 2003). Firms vary in the extent to which they can screen, evaluate and assimilate external inputs to the innovation process. Empirical research underscore that there are substantial variations in the degree to which firms use external ideas (Laursen and Salter, 2006a). Research has shown that firms need competencies in areas related to their partners' to assimilate and co-develop ideas that originate from external sources (Brusoni et al., 2001; Granstrand et al., 1997; Mowery et al., 1996). Internal capabilities and external relations are therefore complements rather than substitutes. Firms spend considerable time and resources on internal R&D, and this leads to the question of what is the right balance between internal and external sources of innovation. What might be new is that the balance between internal and external resources and whether this has shifted, not least because possessing internal R&D capabilities can be argued to be more important when relying heavily on relationships with other actors (Helfat, 2006). This calls for a greater understanding of different forms and practices of openness.

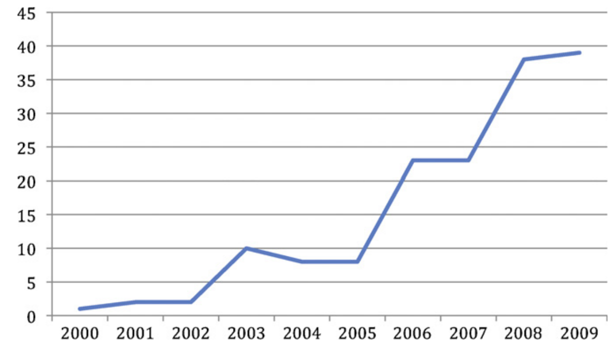

Fig. 1. Number of papers published on open innovation over time.
Source: Own database constructed from ISI data.

Table 1
Most common outlet journals.

Journal name	Number of publications
R&D Management	25
Research Policy	19
Research Technology Management	14
Management Science	9
Industrial and Corporate Change	7
Organization Science	6
Technovation	5
Industry and Innovation	5
International Journal of Technology Management	4
California Management Review	4
MIT Sloan Management Review	4

4. Different types of openness

We analyzed our database to portray the broad trends of research on openness. Fig. 1 shows the number of publications per year, indicating growth in the topic in recent years. This is accentuated by special issues in journals such as *R&D Management, Industry and Innovation* and related special issues on open source, in *Management Science* and *Research Policy*. Indeed *R&D Management* is top, followed by *Research Policy* when it comes to the number of papers published on open innovation. The top ten journals by number of open innovation article published are listed in Table 1.

We continued by investigating the references included in the papers on open innovation. This approach develops from a long tradition in the sociology of science, of references as hallmarks of the intellectual content (Merton, 1973). By investigating what these papers cite, we therefore obtain a richer picture of how this stream of literature links into a broader debate in management and innovation studies. In Table 2, we illustrate the top ten cited references from the papers published on open innovation.

Many of the works on this list are from the open innovation or user innovation literature with von Hippel's (1988) book on top.[5] Unsurprisingly, Chesbrough's (2003a) book is among the most cited references. Some of the most cited works are from scholars not necessarily using an open innovation lens, including Cohen and Levinthal's (1990) work on absorptive capacity, Teece's paper on complementary assets (1986) and March's (1991) exploration and exploitation piece. In analyzing all citations in the work published on open innovation, it is clear that it links into broader debates in innovation studies and management.

To further develop our understanding, we used our database to construct a network of which scholars are collaborating with

[4] See Powell (1990) for a discussion about network arrangements and West and Lakhani (2008) for a distinction between individuals and organization in the case of open innovation.

[5] Although books are not listed as an actual reference in ISI, the references in *cited* papers are included. This is why e.g. von Hippel's books are included.

Table 2
Most cited works of open innovation papers.

Cited paper	Type of publication	Number of cites
von Hippel, E., 1988. The Sources of Innovation. Oxford University Press, New York.	Book	37
Lerner, J., Tirole, J., 2002. Some simple economics of open source. Journal of Industrial Economics, 52, 197–234.	Paper	35
Cohen, W.M., Levinthal, D.A., 1990. Absorptive capacity: a new perspective on learning and innovation. Administrative Science Quarterly, 35(1), 128–152.	Paper	33
Lakhani, K.R., von Hippel, E., 2003. How open source software works: "free" user-to-user assistance. Research Policy, 32(6), 923–943.	Paper	30
von Hippel, E., von Krogh, G., 2003. Open source software and the 'private–collective' innovation model: issues for organization science. Organization Science, 14(2), 209–223.	Paper	30
Hertel, G., Niedner, S., Herrmann, S., 2003. Motivation of software developers in Open Source projects: an Internet-based survey of contributors to the Linux kernel. Research Policy 32 1159–1177.	Paper	29
von Hippel, E., 2005. Democratizing Innovation. The MIT Press, Cambridge, MA.	Book	26
Teece, D., 1986. Profiting from technological innovation: implications for integration collaboration, licensing and public policy. Research Policy, 15, 285–305.	Paper	25
Chesbrough, H.W., 2003. Open Innovation: The New Imperative for Creating and Profiting from Technology. Harvard Business School Press, Boston, MA.	Book	25
March, J., 1991. Exploration and exploitation in organizational learning. Organization Science, 2(1), 71–87.	Paper	24
Chesbrough, H.W., 2003. The era of open innovation. MIT Sloan Management Review Spring, 35–41.	Paper	21
Laursen, K., Salter, A.J., 2006. Open for innovation: the role of openness in explaining innovation performance among UK manufacturing firms. Strategic Management Journal, 27, 131–150.	Paper	20

whom. Fig. 2 illustrates all individuals that have published a paper on open innovation. Nodes are individuals and ties are co-authorships on a publication. We normalized nodes by the number of papers they had published. For instance, Lichtentaler (the most prolific scholar on the topic in terms of number of publications) is the largest node in the network.

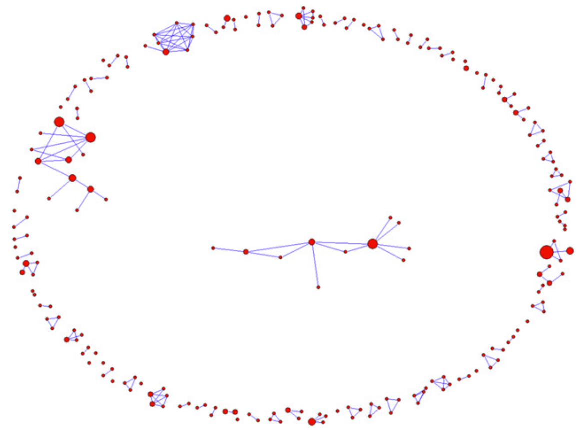

Fig. 2. Illustration of the open innovation community.
Note: Nodes = individuals. Size of nodes = normalized by the number of publications. Ties = co-authorships of publications. Illustrated using the Fruchterman-Reingold algorithm.

In all, 244 scholars have worked on 150 papers. This figure illustrates that the community is relatively fragmented with a few scholars that have collaborated with several others. There are few bridges connecting teams of researchers with the exception of West/Lakhani, who have connected open innovation researchers with scholars investigating user aspects of open innovation. Another exception is Gassmann/Chesbrough accounting for the largest connected component.

We categorized the papers in our database using the distinction of inbound and outbound innovation as a starting point. We further divided inbound and outbound innovation to interactions that are pecuniary versus non-pecuniary and proposed the four different categories illustrated in Table 3. We discuss two different types of inbound innovation—Acquiring and Sourcing, as well as two forms of outbound—Selling and Revealing. We use this analytical framework to structure the review.

We have illustrated that research on openness has increased and deepened in recent years. Our reading of the papers indicates that multiple methods have been used including interviews, case studies and large-scale surveys. Table 4 summarizes a set of empirical work on open innovation with regard to context, sample, key findings and the overall focus of the study. This table is not exhaustive, but it illustrates that different authors conceptualize openness in different ways. Table 4 underscores that large-scale quantitative studies were until recently relatively rare (Laursen and Salter, 2006; Van de Vrande et al., 2009 are recent noteworthy examples of large-scale studies) and that much of the evidence of different types of openness relies on case studies.

We investigated how different papers define openness and how this was conceptualized in empirical investigations. Table 4 shows that there are different types of openness referred to in the literature. We found that while authors discuss openness, it is often unclear exactly what type of openness they were referring to.

Researchers of openness have long argued the benefits of an open approach. However, they have also realized that openness is not a binary classification of open versus closed (Chesbrough, 2003a). The idea behind openness therefore needs to be placed on a

Table 3
Structure of our different forms of openness.

	Inbound innovation	Outbound innovation
Pecuniary	Acquiring	Selling
Non-pecuniary	Sourcing	Revealing

Table 4
Some examples of empirical studies on openness.

Study	Context	Sample	Key results	Focus
Chesbrough and Crowther (2006)	Low-tech or mature industries	12 firms in low-tech or mature industries	Open innovation practices common also in low-tech industries. Leveraging external research as a complement rather than as substitute for internal R&D	Inbound—acquiring
Christensen et al. (2005)	Consumer electronics	Current transformation of sound amplification from linear solid state technology to switched or digital technology within the consumer electronics system of innovation	Different use of open innovation practices is contingent upon the position in the innovation system and stage of the technological regime	Inbound—acquiring
Laursen and Salter (2006)	Firms within manufacturing	2707 firms in the Community Innovation Survey in the UK	Firms external search strategy (breadth and depth) is curvilinearly related to innovative performance	Inbound—sourcing
Fey and Birkinshaw (2005)	Firms with R&D activities	R&D activities of 107 large firms based in the UK and Sweden	How the choice of governance mode for external R&D, along with openness to new ideas and codifiability of knowledge, affects R&D performance	Inbound—sourcing
Henkel (2006)	Embedded Linux	268 developers working with embedded Linux	Firms selectively reveal some technologies to the public as they attach different values to it	Outbound—revealing
West (2003)	Proprietary platform vendors	Three case studies of Apple, IBM and Sun	Proprietary platform firms support open source technologies as part of their platform strategies by balancing the tension between appropriation and appropriability	Outbound—revealing
Lichtenthaler and Ernst (2009)	Multiple industries	155 medium and large-sized firms Germany, Austria and Switzerland	The strategy process and content characteristics jointly shape the performance of out-licensing	Outbound—licensing
Lichtenthaler and Ernst (2007)	Multiple industries	154 medium-sized and large European firms	External technology commercialization is not fully leveraged, but have great potential if successfully implemented	Outbound—licensing

continuum, ranging from closed to open, covering varying degrees of openness. More generally scholars have recognized that some aspects of the innovation process are open and others may be closed (Chesbrough et al., 2006). If we accept that openness is a continuum, a non-controversial argument in the open innovation community, then we can seek to advance a greater understanding of benefits *and* costs of openness. Without considering the disadvantages, the literature is imbalanced and has not leveraged its full potential (Foss, 2003).

4.1. *Revealing: Outbound innovation—non-pecuniary*

4.1.1. *Definition*

This type of openness refers to how internal resources are revealed to the external environment. In particular, this approach deals with how firms reveal internal resources without immediate financial rewards, seeking indirect benefits to the focal firm.

4.1.2. *Advantages*

Allen (1983) used the case of the iron production industry in 19th century England to illuminate what he called collective invention. Firms in this industry regularly shared their designs and the performance of the blast furnaces they had built in verbal interactions and in published material. In most instances, new ideas were not protected by patents, and competing firms could use the information when building new plants. The ability to build upon each others' work resulted in a steady stream of incremental innovation across the community of firms. A detailed case study of innovation in the Cornish mining district during the industrial revolution also shows that firms revealed their internal findings to their competitors (Nuvolari, 2004).

The appropriability regime governs an innovator's ability to capture the profits generated by an innovation (Teece, 1986). Firms usually adopt both formal methods (such as patent, trademark or copyright protection) as well as informal methods (lead times, first mover advantages, lock-ins) within their appropriability strategies. The premise is that openness, caused by voluntarily or unintentionally divulging information to outsiders, does not always reduce the probability of being successful (von Hippel, 1988, 2005; Henkel, 2006; von Hippel and von Krogh, 2003). Henkel (2006), for instance, suggests that firms adopt strategies to *selectively* reveal some of their technologies to the public in order to elicit collaboration, but without any contractual guarantees of obtaining it.

In the absence of strong IPR, in some cases there are greater chances of cumulative advancements (Levin et al., 1987). This contention is highlighted in the existence of phenomena such as *Wikipedia* and free and open source software where individuals collectively develop innovative solutions (West and Gallagher, 2006). In the literature on standards, for example, it is well-known that being open and focusing less on ownership increases the opportunities to gain interest from other parties.

It has been found that firms can place too strong an emphasis on protecting their knowledge, resulting in a 'myopia of protectiveness' (Laursen and Salter, 2006b). In this case, firms may become obsessed with ownership, instead of marshalling resources and support from the external environment necessary to bring inventions into commercial applications and services. Firms therefore use combinations of different means of protection, balancing the relative inefficiency of formal protection by putting greater emphasis on alternative methods (López and Roberts, 2002).

4.1.3. Disadvantages

An obvious disadvantage of revealing internal resources to pace the general technological advance is the difficulty in capturing benefits that accrue (Helfat, 2006). Competitors can be better positioned with complementary assets and production facilities to make use of the technological advance.

It is a challenge to choose what internal resources to reveal to the external environment. Some large companies have different committees that make decisions whether to file patents or disclose. Smaller companies, in contrast, typically lack the resources to structure this process.

4.2. Selling: outbound innovation—pecuniary

4.2.1. Definition

This type of openness refers to how firms commercialize their inventions and technologies through selling or licensing out resources developed in other organizations.

4.2.2. Advantages

Chesbrough (2003a,b, 2006) discusses how firms can benefit by commercializing inventions by selling or licensing-out ideas that might hitherto have been ignored. Some firms have developed a plethora of patents because of incentives used in R&D to encourage patenting—often without considering business relevance (Nerkar, 2007). By selling or out-licensing, firms can more fully leverage their investments in R&D, partnering with actors adept at bringing inventions to the market place. Chesbrough and Rosenbloom (2002, p. 550) note that . . . "if companies that fund research that generates spill-overs are to develop a better business model to commercialize their spill-over technologies, our traditional notions of technology management must be expanded". Gassmann and Enkel (2006) discuss how some firms adopt different 'inside–out' processes to externalize internal knowledge and invention to the market place.

Research suggests that licensing out inventions and technologies is becoming more common. Some firms have even made it a strategic priority to out-license technologies and inventions (Fosfuri, 2006). Indeed, there are some success stories portrayed in the literature, but there are often many obstacles that prevent firms from selling or licensing-out technologies (Rivette and Kline, 2000).

4.2.3. Disadvantages

Market failure sometimes occurs because inventors are reluctant to reveal their developments. Arrow's (1962) seminal paper suggests the significant challenge involved in reaching agreements based on information, when two or more parties are involved. When an inventor is keen to license its information to a potential licensee, it is necessary to reveal some information to the potential customer. This 'disclosure paradox' implies that the potential licensee receives the information without paying for it and could – in principle – act opportunistically and steal the idea. Arrow argued that such problems cause market failures because they make inventors reluctant to reveal their technology or knowledge.

The market for technology literature has argued that there are significant transaction costs involved in transferring technologies between organizations. As a consequence, the potential of selling technologies in the market place has not been fully leveraged (Gambardella, Giuri, & Luzzi, 2007). Gambardella et al. (2007) even suggest that the market for technology could be close to 70% larger should some obstacles be overcome.

An obstacle that often prevents firms from out-licensing technologies is that they have difficulty anticipating the potential value (Chesbrough and Rosenbloom, 2002). Firms may be over-committed to where they have invested resources, another organization may be better equipped to independently commercialize it. Chesbrough and Rosenbloom's (2002) analysis of Xerox illustrate how the combined market capitalization of spin-offs and other external commercializations subsequently overtook the value of Xerox. With this potential, it is clear that a deliberate strategy may need to be in place. Lichtenthaler and Ernst (2007) suggest that while many firms are open to licensing technologies, they lack a conscious strategy for bringing this into practice. In a recent paper, Lichtenthaler (2009) discussed how strategic planning and the content characteristics jointly shape the firms potential to benefit from out-licensing.

In many technology environments, patents provide opportunities for firms to overcome this disclosure problem (Arrow, 1962). Open innovation requires that buyer and seller reach an agreement, so appropriability regimes allow the seller to disclose information. The contention is thus that IP is important in order to trade innovations (West, 2006). Understanding the disclosure paradox calls attention to the means of appropriability in open innovation, and how firms attempt to be open yet are able to appropriate commercial returns from their innovative efforts. To overcome this paradox, firms often require that inventors have formal IPR in place before they work together.

4.3. Sourcing: inbound innovation—non-pecuniary

4.3.1. Definition

This type of openness refers to how firms can use external sources of innovation. Chesbrough et al. (2006) claim that firms scan the external environment prior to initiating internal R&D work. If existing ideas and technologies are available, the firms use them. Accounts of corporate R&D laboratories show that they are vehicles for absorbing external ideas and mechanisms to assess, internalize and make them fit with internal processes (Freeman, 1974).

4.3.2. Advantages

The SAPPHO project showed that firms rely on many external sources of ideas (Rothwell et al., 1974). According to Rothwell (1994, p. 19. . . "accessing external know-how has long been acknowledged as a significant factor in successful innovation". Following this tradition of research, Laursen and Salter (2004, p. 1204) define openness as "the number of different sources of external knowledge that each firm draws upon in its innovative activities". Their logic is that the larger the number of external sources of innovation, the more open will be the firm's search strategy. This is highlighted in much other open innovation literature, which underlines that innovation is often about leveraging the discoveries of others.

Firms that manage to create a synergy between their own processes and externally available ideas may be able to benefit from the creative ideas of outsiders to generate profitable new products and services. Available resources become larger than a single firm can manage, they enable innovative ways to market, or the creation of standards in emerging markets. Such synergies can be created by relying on the external environment, and by taking an active part in external developments. P&G (Procter and Gamble), for example, have been able to realize value from new products that they developed and marketed jointly with their re-seller network, when hitherto they would not have thought to leverage these complementary assets in order to bring products to market (Huston and Sakkab, 2006). Leiponen and Helfat (2005) analyzed the Finnish Community Innovation Survey and discovered positive implications for success through following a 'parallel-path strategy in innovation', where firms maintain an open strategy of sourcing information (breadth in sources) together with an 'open mind' about the paths to innovation (breadth of objectives).

4.3.3. Disadvantages

Some scholars have begun to stress potential limitations. There may be a cultural difference here, too, with Japanese firms, for example, Toyota and Canon, employing carefully crafted search strategies yet managing their internal R&D facilities in a relatively closed manner.

Individuals can only focus properly on a few tasks at any point in time—the span of attention problem (Simon, 1947). There are cognitive limits to how much we can understand. Some organizations over-search by spending too much time looking for external sources of innovation. Building on this reasoning, Katila and Ahuja (2002) propose that search behaviour is critical for understanding the limits and contingent effects on innovation. Based on a study of the industrial robotics industry, they suggest that some firms over-search and that there is thus a curvilinear relationship between innovative performance and their search for new innovations. Laursen and Salter (2006a) extend this reasoning by looking at external sources of innovation. Drawing on a sample from a large UK survey of 2707 manufacturing firms, they show that wide and deep search for sources of innovation is curvilinearly related to innovative performance. In other words, while there may be an initial positive effect on openness, firms can over-search or come to rely too heavily on external sources of innovation.

It is not obvious that all firms will rely on external partners. In fact, there are substantial variations in the degree to which firms adopt open innovation (Laursen and Salter, 2004), and the degree of openness varies according to external sources of innovation as technologies mature (Christensen et al., 2005). In this regard, research could benefit from concentrating more explicitly on the particular nature and context of external sources of innovation (Gassmann, 2006).

4.4. Acquiring: Inbound innovation—pecuniary

4.4.1. Definition

This type of openness refers to acquiring input to the innovation process through the market place. Following this reasoning, openness can be understood as how firms license-in and acquire expertise from outside.

4.4.2. Advantages

Acquiring valuable resources to an innovation process requires expertise. While acknowledging the importance of openness in terms of external sources of innovation, von Zedtwitz and Gassmann (2002) state that in order to invest when openness is high, firms need some degree of control over a number of the elements in their networks. Although there are many benefits from being able to buy or in-source external ideas to the organization, expertise is required to search for and evaluate them.

4.4.3. Disadvantages

A further point relates to the similarity of knowledge bases and how they facilitate the integration of ideas from distant realms (Kogut and Zander, 1992), because shared languages, common norms and cognitive configurations enable communication (Cohen and Levinthal, 1990). Incorporating knowledge bases too close to what the firm already knows may hamper the positive effect of assimilating external inputs. For instance, Ahuja and Katila (2001) suggest that knowledge relatedness between the acquiring and acquired firms is curvilinearly related to innovative performance. Too distant inputs are harder to align with existing practices, and if knowledge bases are too similar it is difficult to come up with novel combinations (Sapienza et al., 2004). In other words, the effectiveness of openness is contingent upon the resource endowments of the partnering organization.

4.5. Combining different types of openness

We have argued that there are different forms of openness and exemplified the benefits and disadvantage of each. In Table 5, we summarize the types of openness with regard to the main focus and the logic behind the degree of openness.

In reading through the 150 papers in our database, we found that most papers analyze one or two different forms of openness. Until quite recently, there have been few systematic attempts to investigate several different forms of openness (with the noteworthy exceptions of Acha, 2007; Van de Vrande et al., 2009). A largely unattended research area is how different forms of openness can be combined. Chesbrough and Rosenbloom (2002) have noted that a critical option for firms is to choose between different forms of openness in developing the firm's business model. In the discussion below, we note how one fruitful area for future studies is to investigate how different forms of openness jointly shape performance.

5. Discussion

We began this paper with the observation that in spite of the growing literature on openness, there is a lack of clarity and some dissatisfaction with how the concept has been used. The obvious disadvantage of this is that comparing empirical findings is difficult because the literature is fragmented. We therefore set out to explore the foundations of the literature and further define different types of openness. We divided inbound and outbound innovation to pecuniary and non-pecuniary interactions as illustrated in Table 3. For each type of openness, we explored the advantages as well as disadvantages. In this section, we elaborate on the implications of this exercise for theory and practice.

5.1. Implications for theory

West et al. (2006) developed a research agenda for open innovation based on observations from industry practices firms adopt to rely upon external actors. A review of the earlier work makes it clear that innovation has to some extent always been open. Empirically, practices to rely upon external actors in the innovation process have been around for decades. For instance, Hargadon's (2003) work illustrate that firms have always relied on outflows and inflows of ideas, resources and individuals and Freeman (1974) proposed that R&D laboratories are not 'castles on the hill'. Theoretically, researchers have analyzed the benefits that can accrue from involvement in various kinds of partnerships. The open innovation literature synthesized these ideas and stimulated a renewed interest from scholars and policy makers alike.

If progress is to be made in research on the changing nature of innovation processes, and that if firms are to develop viable strategies for innovation management, more precision is needed in conceptualizing open innovation. Chesbrough et al. (2006, p. 286) submit that "open innovation is both a set of practices for profiting from innovation, and also a cognitive model for creating, interpreting and researching those practices". This definition is inclusive for various different practices to be considered open. An unintended consequence of such a definition is that scholars

Table 5
Comparison of four different types of openness.

	Outbound innovation Revealing	Outbound innovation Selling	Inbound innovation Sourcing	Inbound innovation Acquiring
Logic of exchange	Non-pecuniary—indirect benefits	Pecuniary—money involved in exchange	Non-pecuniary—indirect benefits	Pecuniary—money involved in exchange
Focus	Revealing internal resources to the external environment (e.g. Allen, 1983; Henkel, 2006; Nuvolari, 2004; von Hippel and von Krogh, 2003)	Out-licensing or selling products in the market place (e.g. Lichtenthaler and Ernst, 2009; Chesbrough and Rosenbloom, 2002)	Sourcing external ideas and knowledge from suppliers, customers, competitors, consultants, universities, public research organizations, etc. (e.g. Fey and Birkinshaw, 2005; Lakhani et al., 2006; Laursen and Salter, 2006a)	Acquiring inventions and input to the innovative process through informal and formal relationships (e.g. Chesbrough and Crowther, 2006; Christensen et al., 2005)
Advantages and disadvantages shaping extent of openness				
Advantages driving openness	Marshal resources and support (Henkel, 2006)	Commercialize products that are 'on the shelf'	Access to a wide array of ideas and knowledge (Laursen and Salter, 2006a)	Gaining access to resources and knowledge of partners (Powell et al., 1996)
	Gaining legitimacy from external environment (Nuvolari, 2004)	Outside partners may be better equipped to commercialize inventions to the mutual interests of both organizations (Chesbrough and Rosenbloom, 2002)	Discovering radical new solutions to solving problems (Lakhani et al., 2006)	Leveraging complementarities with partners (Dyer and Singh, 1998)
	Foster incremental and cumulative innovation (Murray and O'Mahony, 2007; Scotchmer, 1991)			
Disadvantages driving closeness	Difficult to capture the benefits that accrue	Over-commitment to own product and technologies make it difficult to out-license (Lichtenthaler and Ernst, 2007)	Many sources create an attention problem (Laursen and Salter, 2006a)	Difficult to maintain a large number of ties with different partners (Ahuja, 2000)
	Internal resources can leak to competitors (Laursen and Salter, 2006b)		Difficult to choose and combine between too many alternatives (Sapienza et al., 2004)	Risk of outsourcing critical dimension of the firm's business

use the openness construct to mean different things. Scholars have used different definitions of openness in their studies of 'open innovation'. This has led to conceptual ambiguity, with empirical papers focusing on different aspects, inhibiting our ability to build a coherent body of knowledge. At the very least, this may be detrimental to what could be useful studies by scholars outside the innovation community, who find it difficult to consider how openness links into broader debates. This is supported by a recent study by Acha (2007) that uses the Community Innovation Survey in the UK to test openness in different ways. She notes that it is not clear how or whether different types of openness are correlated in empirical studies, further highlighting the need to consider different categories of openness. Building upon much earlier literature in the field, our paper advanced a systematic framework that critically analyzed this growing literature.

Work on openness has used case studies of large American technology companies such as Lucent, Intel, 3Com and Millenium Pharmaceuticals (Chesbrough, 2003a). However, exploring other empirical settings is important for achieving external validity (Chesbrough and Crowther, 2006). Until recently, few studies explored openness using large-scale datasets covering a variety of different industries. With a growing number of scholars penetrating the field, we have witnessed a change in the type of methods employed (see e.g. Laursen and Salter, 2006a; Lichtenthaler and Ernst, 2009; Van de Vrande et al., 2009). These studies complement case studies that provide rich process-oriented descriptions of openness that in isolation might risk becoming anecdotal.

Why are some firms profiting more than others from openness? This is a fundamental question on which there is surprisingly limited evidence. Researchers on openness have suggested that it may be necessary to keep some aspects of the innovation process open while others remain closed (Laursen and Salter, 2006a; von Zedtwitz and Gassmann, 2002). Almost all the published papers on this subject focus on the potential benefits of openness, without theorizing about the disadvantages. Mowery (1983) suggested that internal R&D laboratories emerged because it was more efficient to internalize R&D compared to buying in the market place. It may be that some of the forces explained in the literature on openness have eradicated these costs. In sum, at least two different forms of cost emerge when collaborating with external partners–the costs of coordination and competition (Grant, 1996). Costs of coordination emerge from organizations that are different, where it may be difficult to bridge organizational boundaries. In the openness literature, maintaining too many relationships is costly and may lead to a diversion of managerial attention. There are also costs of competition that emerge from the risk that one actor would act opportunistically in bad faith. Another potential cost is related to protecting ideas to which others have access. Our paper provides a framework to analyze advantages and disadvantages of different forms of openness. This helps to understand when advantages outweigh the disadvantages of openness.

A number of contingent factors explain why some firms are better equipped to profit from openness. For example, some scholars suggest that stronger IPR regimes are associated with a higher reliance on external actors. Gallini (2002) predicted that stronger IPR promote greater willingness to license as protection for ideas allow organizations to respond to opportunistic behaviour with legal sanctions. This idea is endorsed by Laursen and Salter (2006b) who showed that reliance on many external sources of innovation is higher in industries with formal protection for ideas.

5.2. Implications for practice

Open innovation has been influential and appears to be adopted by many organizations (Chesbrough and Crowther, 2006; Gassmann and Enkel, 2006). But openness can be costly (Laursen and Salter, 2006a). Open innovation and the business models relating to it are changing practice and creating strategic positions that hitherto had not been clearly articulated (Chesbrough and Appleyard, 2007). The success of open innovation can differ across technologies and industries (Christensen et al., 2005). It is therefore important to attend to the barriers and limits to bringing credible and useful insights to practitioners.

There can often be difficulties in evaluating external rather than internal ideas, as there is much less first-hand information available on external ideas (Menon and Pfeffer, 2003). The capabilities required to work as a 'broker', recombining ideas from inside and outside the firm are likely to be different from those found in traditional internal R&D settings. Being more involved in open innovation can therefore create tensions with other practices within the organization. This begs the question of how firms might operationalize strategies to enable them to benefit from more open approaches, what mechanisms might be implemented and how resources and capabilities should be deployed to support them.

5.3. Future research

There are several areas for future research that emerge from our paper.

First, we have found that there is a limited understanding about the costs of openness. For each form of openness, we considered the associated benefits and costs. An expansion of evidence on these arguments will be useful to explain under which contingencies openness is a fruitful strategy.

Second, while firms have always relied on some degree of openness, there may be new ways to work with external actors, suggesting qualitative change of practices. As we have demonstrated, empirical work using qualitative and quantitative approaches to analyze open innovation processes has begun to emerge. It is important to note the risks of being pre-occupied with exploring the optimal level of openness rather than probing how openness has changed in a qualitative sense. Perhaps openness is today taking different forms than in the past, particularly given the availability of new information and communication infrastructures to support innovation, what some call 'innovation technology' (Dodgson et al., 2005). These technologies allow for ideas to be worked on, exchanged and diffused rapidly with decreased transmission costs and a larger potential range and number of participants. Early research explored the impact of electronic tool-kits on innovation process, and later work suggests that the emergence of innovation technologies leads to an intensification of innovation and reduced uncertainty of outcomes (Dodgson et al., 2005). Research on the relationship between these information and communication technologies and strategies benefits and disadvantages of openness might provide fruitful insights for theory and practice. Firms adopt new practices to cope with openness and create competitive businesses. In this regard, research could benefit from concentrating more explicitly on the particular nature and context of external sources of innovation (Gassmann, 2006). Implicitly, external knowledge is considered to be 'out there' ready to be harnessed by firms, and we have limited understanding of the process of sourcing this into corporations.

Third, while there are many descriptions of how external relations influence performance, there is less research focused on the underlying decision process, which is important as firms face difficulties in maintaining large number of relations. It might be expected that the search process would experience some inertia and consistent patterns of collaboration over time due to socialization. Rather than hedge against risk and avoid redundancy by diverse relations, there can be persistent pathways to source knowledge. Inertia in the search processes for external expertise could result in continuation of relations that are not meeting expectations producing over-embeddedness (Uzzi, 1997), with too little diversity in terms of partners, which raises another set of questions for empirical exploration.

Fourth, further research would appear to be necessary to elaborate on the conceptual frame for open innovation from the perspective of product/technology lifecycles and the different phases through which an innovation evolves from conceptualization to commercialization. This will require a technology-analysis lens, which was out of the scope of this paper.

Fifth, future research could elaborate further on the different combinations of openness. A few scholars have looked at different governance modes (van de Vrande et al., 2006; Fey and Birkinshaw, 2005), but we lack substantive evidence about how firms can combine different ways of managing openness. van de Vrande et al's (2009) paper highlight various forms of openness, and we see it as a natural next step to investigate conditions when different forms of openness complement or substitute one another. Papers in this vein might advance both a greater theoretical story, and managerial practice.

6. Concluding remark

A number of papers draw on the openness construct to depict and explain different aspects of the innovation process. It is timely to take stock of this important idea and assess whether a coherent body of literature is being developed. This paper presents reservations about the definition of openness and the ways in which the construct is used in a variety of empirical research settings. In reviewing the results of this research we have identified different interpretations of the openness construct and provided a conceptual frame for what we hope is a more coherent means of developing the research agenda. We hope that this paper will inspire researchers to build on the concept and theorize in a way that helps explain why some firms profit more than others from openness.

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